

Contact

2672999245 (Mobile)
karthikheyaa@gmail.com

www.linkedin.com/in/karthikheyaa
(LinkedIn)

Top Skills

PyTorch
Scientific Computing
Optimization

Certifications

Mathematics for Machine Learning:
Linear Algebra
TensorFlow Developer Certificate
AWS Certified Machine Learning
Engineer – Associate

Karthikheyaa Kurra

Research Associate at University of Delaware | Department of
Mathematical Sciences | Scientific Machine Learning
Newark, Delaware, United States

Summary

I do research in scientific machine learning at the University of Delaware. The problem I work on: classical PDE solvers are iterative, and every new problem restarts the iteration from scratch. With Dr. Ke Chen, I'm building neural surrogates that learn the solution map for semilinear elliptic PDEs, so a new problem gets solved in a single forward pass instead. This matters in many-query settings like design optimization and uncertainty quantification, where the same family of problems gets solved thousands of times. A manuscript is in preparation, so the specifics stay off the internet for now. Most of my actual day is the unglamorous side of research: generating high-precision ground truth with spectral discretizations and Newton solvers, running controlled ablations, profiling inference, and keeping the codebase trustworthy with tests. I also work with Dr. Sunita Chandrasekaran's group benchmarking LLM inference and serving on NVIDIA DGX hardware, for an on-premise assistant the university is building. I came into this from a computer science background (B.E. in CSE, then an M.S. in Data Science at UD). The engineering comes naturally; the math I'm rebuilding deliberately, one textbook at a time. I write about what I'm learning at khey17.github.io.

Experience

University of Delaware

1 year 2 months

Research Associate

June 2026 - Present (2 months)

Newark, DE

(1) Scientific machine learning with Dr. Ke Chen (Dept. of Mathematical Sciences): continuing our work on learned surrogates for semilinear elliptic PDE solvers. I own the research infrastructure end to end: data generation with spectral discretizations and Newton solvers, PyTorch training and evaluation

pipelines, ablation studies, inference profiling, and a regression-locked, unit-tested codebase. (Work in progress).

(2) LLM serving with Dr. Sunita Chandrasekaran's group (Dept. of Computer & Information Sciences): benchmarking LLM inference and serving performance (throughput, latency, model selection) on NVIDIA DGX hardware for an on-premise assistant serving University of Delaware students, faculty, and staff.

Research Assistant

June 2025 - June 2026 (1 year 1 month)

Newark, DE

Research assistant with Dr. Ke Chen (Dept. of Mathematical Sciences), working on neural surrogates for nonlinear PDE solvers: replacing iterative Newton solves with learned models that solve new problem instances in a single forward pass. Built the project's ground-truth generation (Chebyshev spectral discretizations + Newton iteration), dataset validation, and reproducible PyTorch training and evaluation pipelines; drove architecture and loss design through controlled ablation studies.

Indian Institute of Technology, Hyderabad

Research Intern

March 2023 - June 2023 (4 months)

Hyderabad

Case study applying supervised contrastive learning (Khosla et al., 2020) to facial expression recognition, comparing SupCon loss against a standard cross-entropy baseline in PyTorch with face detection and data augmentation. Analyzed learned embeddings with t-SNE to diagnose class separation issues and delivered a reproducible codebase to the lab.

Sathyabama Institute of Science & Technology, Chennai

Teaching Assistant

January 2023 - April 2023 (4 months)

Chennai

Teaching assistant for the undergraduate Machine Learning course.

The University of Texas at Dallas

Summer Intern

June 2022 - July 2022 (2 months)

Dallas, TX

Building Machine Learning models from scratch (Understanding the algorithms and the mathematical idea behind them)

Education

University of Delaware

Master's degree, Data Science · (August 2024 - May 2026)

Sathyabama Institute of Science & Technology, Chennai

Bachelor of Engineering - BE, Computer Science · (October 2020 - May 2024)